Synopsis

Olympic Games Performance Prediction using Machine Learning

Problem Id : 26

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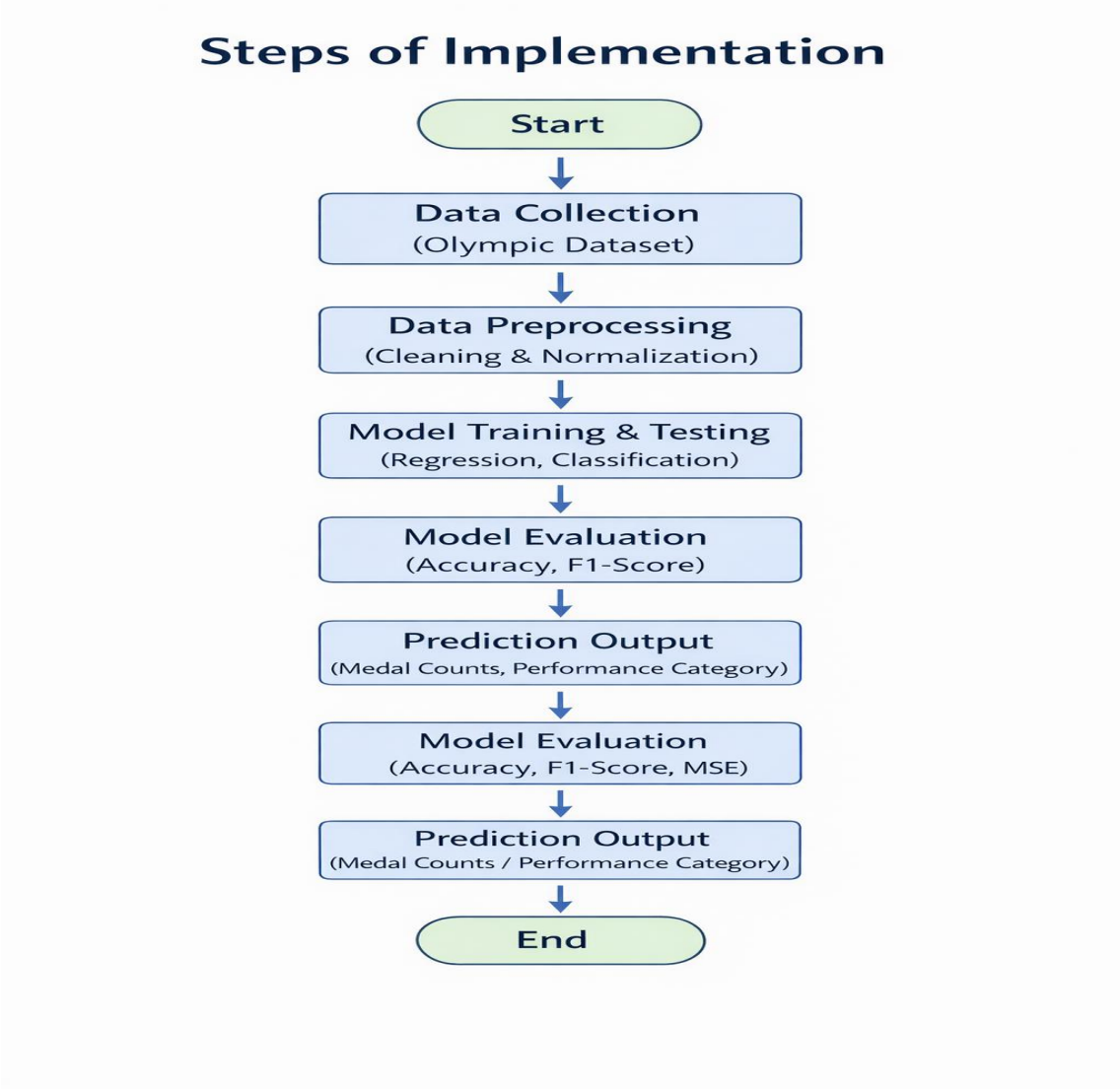
- **Problem Description:**

This project aims to design an end-to-end machine learning system to analyze and predict Olympic Games performance using numerical sports data. The dataset includes features such as athlete participation, medal counts, event statistics, and country-wise performance over multiple years. The goal is to identify patterns and trends in historical Olympic data and use them to predict future medal outcomes.

In real life, this system can help sports authorities and governments make better decisions regarding training, funding, and athlete selection. It can also help analysts understand which factors (like participation or past performance) most influence medal counts. The project involves data preprocessing, exploratory data analysis (EDA), feature selection, model training, and evaluation. Both regression and classification techniques will be used to predict medal counts or categorize country performance levels. This project demonstrates how machine learning can be applied in sports analytics for performance improvement and strategic planning.

- **Steps of Implementation:**

- **Data Collection**
  Collect Olympic dataset from reliable sources like Kaggle.
- **Data Preprocessing**
  Clean the data by handling missing values, removing duplicates, and formatting.
- **Model Selection**
  Choose suitable machine learning models (Regression/Classification).
- **Training and Validation**
  Train the model using training data and validate its performance.
- **Evaluation**
  Evaluate the model using metrics like accuracy, F1-score, or MSE.
- **Final Model & Prediction**
  Generate the final model and use it to predict Olympic performance.



- **Expected Modules/ Libraries to be used :**
- **NumPy** – For numerical operations
- **Pandas** – Data handling and preprocessing
- **Matplotlib** – Data visualization (graphs)
- **Seaborn** – Advanced visualization
- **Scikit-learn** – ML models and evaluation
- **Jupyter Notebook** – Development environment

- **Expected Evaluation metrics and Plots to be used:**

- **Accuracy** – For classification performance
- **Precision & Recall** – For detailed evaluation
- **F1-Score** – Balance between precision & recall
- **Mean Squared Error (MSE)** – For regression
- **Confusion Matrix** – To analyze predictions
- **Plots:**
    - Bar chart (medal counts)
    - Line graph (performance over years)
    - Heatmap (correlation)

- ## Required topics from the subject:
- **Strings** – Used for handling country names, athlete names
- **Pointers** – Memory handling in C (basic understanding)
- **Files** – Reading dataset files (CSV)
- **Arrays** – Storing numerical data
- **Functions** – Modular coding

- ## Platform Required:
    - Jupyter Notebook
    - Python (Anaconda / VS Code)

- ## Books and Link Sources:
- Hands-On Machine Learning – Aurélien Geron
- Python for Data Analysis – Wes McKinney
- Scikit-learn official documentation

- Kaggle (Olympics dataset)
- GeeksforGeeks (ML concepts)